# S1 Text. Supporting methods and numerical results

Electrostatic–hydrophobic cooperation in the mesoscale assembly of synapsin and synaptic vesicles

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## S1. Design and structural observables

The factorial design contains four E/H conditions and three seed blocks, 9310002–9310004. Each trajectory has 200 SVs and 1,807 Syn agents in a spherical compartment of volume $0.1\,\mu\mathrm{m}^3$. All trajectories were followed to 84 ms at a fixed integration step of 0.0007 ms. Van der Waals interactions remain enabled in every condition. The 600 unique saved frames are separated by 0.14 ms. Continuations retain the same physical parameters and saved random state; identical checkpoint frames are counted once.

The principal 84-ms summaries average frames 421–600 (58.94–84 ms) within each trajectory, followed by a mean and sample SD across seeds. The earlier 28-ms summaries use frames 141–200 (19.74–28 ms). Both apply the same final-30% rule. Each trajectory is one independent replicate. The two horizons are repeated observations of the same runs.

The association graph uses $\|x_i - x_j\| \le a_i + a_j + 2\,\mathrm{nm}$, where $a_i$ is the outer radius. $L$ is the all-agent largest-component fraction, $Q_{\mathrm{Syn}}$ is the non-isolated Syn fraction and $L_{\mathrm{SV}}$ is the fraction of all SVs in the component selected by total agent count. Non-isolated Syn includes Syn–Syn groups and Syn–SV associations. The original machine-readable column "synapsin_recruited_fraction" and the updated column "syn_nonisolated_fraction" both mean $Q_{\mathrm{Syn}}$.

## S2. Complete per-run structural observables at 84 ms

| Condition | Seed | $L$ | $Q_{\mathrm{Syn}}$ | $L_{\mathrm{SV}}$ | Drift-limited |
|---|---|---|---|---|---|
| E0H0 | 9310002 | 0.027675 | 0.318434 | 0.084889 | 0.000001 |
| E0H0 | 9310003 | 0.054091 | 0.319935 | 0.171139 | 0.000000 |
| E0H0 | 9310004 | 0.028647 | 0.313272 | 0.086889 | 0.000000 |
| E0H1 | 9310002 | 0.337812 | 0.560004 | 0.545417 | 0.001161 |
| E0H1 | 9310003 | 0.413553 | 0.592714 | 0.587556 | 0.002336 |
| E0H1 | 9310004 | 0.280632 | 0.550593 | 0.435806 | 0.001121 |
| E1H0 | 9310002 | 0.445244 | 0.759017 | 0.484389 | 0.000000 |
| E1H0 | 9310003 | 0.599812 | 0.779850 | 0.616083 | 0.000000 |
| E1H0 | 9310004 | 0.462299 | 0.752702 | 0.496528 | 0.000000 |
| E1H1 | 9310002 | 0.921386 | 0.953849 | 0.908056 | 0.005209 |
| E1H1 | 9310003 | 0.912982 | 0.949324 | 0.868333 | 0.005391 |
| E1H1 | 9310004 | 0.925491 | 0.951593 | 0.896361 | 0.005518 |

The final column gives the whole-84-ms mean fraction of particle updates affected by the deterministic drift limiter, weighted by integration steps across segments. Brownian increments are retained in full. All segments completed with finite positions and no unconverged projection steps at the original 0.005-nm residual tolerance.

## S3. Paired interaction contrasts at both horizons

The contrast is $I_s = Y_{11,s} - Y_{10,s} - Y_{01,s} + Y_{00,s}$. The following table retains every seed and each structural observable.

| Horizon (ms) | Seed | $I_L$ | $I_{Q_{\mathrm{Syn}}}$ | $I_{L_{\mathrm{SV}}}$ |
|---|---|---|---|---|
| 28 | 9310002 | 0.359043 | -0.029266 | 0.380500 |
| 28 | 9310003 | 0.313237 | -0.031452 | 0.307500 |
| 28 | 9310004 | 0.444503 | -0.026075 | 0.455250 |
| 84 | 9310002 | 0.166005 | -0.046738 | -0.036861 |
| 84 | 9310003 | -0.046291 | -0.103305 | -0.164167 |
| 84 | 9310004 | 0.211208 | -0.038431 | 0.050917 |

At 84 ms, the mean $I_L$ is 0.110307, with between-seed SD 0.137488. The approximate 95% paired Student-$t$ interval is [-0.231233, 0.451848].

The main text uses paired differences and sample SDs. There are three seed blocks and two degrees of freedom for paired intervals. Normality of seed differences is difficult to assess with three blocks. The full-precision conditional contrasts are retained in S1 Data. The original 28-ms statistical audit remains a historical analysis of that horizon and is stored separately from the 84-ms tables.

## S4. Association-shell sensitivity at 84 ms

Only the graph threshold changes; trajectories and forces are fixed. Entries are means of three per-run late-window means.

| Condition | Shell (nm) | $L$ | $L_{\mathrm{SV}}$ |
|---|---|---|---|
| E0H0 | 0 | 0.018471 | 0.070759 |
| E0H0 | 2 | 0.036805 | 0.114306 |
| E0H0 | 4 | 0.073231 | 0.193611 |
| E0H0 | 6 | 0.143276 | 0.328630 |
| E0H1 | 0 | 0.257311 | 0.402796 |
| E0H1 | 2 | 0.343999 | 0.522926 |
| E0H1 | 4 | 0.411524 | 0.614833 |
| E0H1 | 6 | 0.473130 | 0.692194 |
| E1H0 | 0 | 0.346878 | 0.363213 |
| E1H0 | 2 | 0.502452 | 0.532333 |
| E1H0 | 4 | 0.596171 | 0.644713 |
| E1H0 | 6 | 0.666221 | 0.730852 |
| E1H1 | 0 | 0.887193 | 0.857648 |
| E1H1 | 2 | 0.919953 | 0.890917 |
| E1H1 | 4 | 0.929835 | 0.909870 |
| E1H1 | 6 | 0.938186 | 0.927667 |

## S5. Species-resolved contact analysis at 84 ms

For each Syn agent $i$, the SV-neighbor count is

$$d_i^{\mathrm{SV}} = \sum_{j \in \mathcal{V}} \mathbf{1}\{\|x_i - x_j\| \le a_i + a_j + 2\,\mathrm{nm}\}.$$

The two Syn fractions count $d_i^{\mathrm{SV}} \ge 1$ and $d_i^{\mathrm{SV}} \ge 2$, respectively, divided by $N_{\mathrm{Syn}}$. Reciprocal SV coverage counts SVs with at least one Syn neighbor, divided by $N_{\mathrm{SV}}$. The same threshold is applied to every condition. Values are averages over 58.94–84 ms within each run.

| Condition | Seed | Syn: $\ge 1$ SV | Syn: $\ge 2$ SV | SV: $\ge 1$ Syn |
|---|---|---|---|---|
| E0H0 | 9310002 | 0.191407 | 0.012384 | 0.805500 |
| E0H0 | 9310003 | 0.191063 | 0.013552 | 0.810472 |
| E0H0 | 9310004 | 0.189341 | 0.012141 | 0.814444 |
| E0H1 | 9310002 | 0.454052 | 0.124909 | 0.957778 |
| E0H1 | 9310003 | 0.501024 | 0.192194 | 0.950778 |
| E0H1 | 9310004 | 0.438929 | 0.119185 | 0.952000 |
| E1H0 | 9310002 | 0.745207 | 0.193448 | 0.993083 |
| E1H0 | 9310003 | 0.768591 | 0.253127 | 0.988639 |
| E1H0 | 9310004 | 0.738695 | 0.182454 | 0.992444 |
| E1H1 | 9310002 | 0.951928 | 0.664957 | 0.984139 |
| E1H1 | 9310003 | 0.947703 | 0.676736 | 0.975222 |
| E1H1 | 9310004 | 0.949920 | 0.664690 | 0.983000 |

Mean unordered edge counts per snapshot:

| Condition | Seed | Syn–Syn | Syn–SV | SV–SV |
|---|---|---|---|---|
| E0H0 | 9310002 | 157.572 | 368.650 | 105.744 |
| E0H0 | 9310003 | 159.033 | 370.122 | 122.639 |
| E0H0 | 9310004 | 151.844 | 364.400 | 102.083 |
| E0H1 | 9310002 | 398.800 | 1069.344 | 179.533 |
| E0H1 | 9310003 | 610.489 | 1321.411 | 208.717 |
| E0H1 | 9310004 | 398.356 | 1031.139 | 164.022 |
| E1H0 | 9310002 | 296.000 | 1696.178 | 28.694 |
| E1H0 | 9310003 | 366.533 | 1846.317 | 39.150 |
| E1H0 | 9310004 | 281.094 | 1664.517 | 27.378 |
| E1H1 | 9310002 | 1331.250 | 2982.600 | 126.700 |
| E1H1 | 9310003 | 1390.100 | 3020.233 | 132.522 |
| E1H1 | 9310004 | 1396.917 | 2953.311 | 125.256 |

## S6. Short-lag mobility and experimental scale reference

Syn positions are projected onto $xy$, converted from nm to $\mu$m, and fitted using per-agent time-averaged MSD with a free intercept over native lags 2.10–3.92 ms (14 lag points). The apparent coefficient is slope/4 in $\mu m^2/s$. Negative fitted slopes are retained. The full 600-frame trajectory is used for the main figure. The final 180 frames provide a window-sensitivity comparison.

| Condition | Seed | $D$: full | $D$: late | Negative: full | Negative: late |
|---|---|---|---|---|---|
| E0H0 | 9310002 | 0.141098 | 0.143295 | 0.0000 | 0.0033 |
| E0H0 | 9310003 | 0.140255 | 0.137813 | 0.0000 | 0.0033 |
| E0H0 | 9310004 | 0.139690 | 0.139443 | 0.0000 | 0.0050 |
| E0H1 | 9310002 | 0.131435 | 0.129700 | 0.0000 | 0.0039 |
| E0H1 | 9310003 | 0.129508 | 0.122338 | 0.0000 | 0.0017 |
| E0H1 | 9310004 | 0.131327 | 0.128424 | 0.0000 | 0.0022 |
| E1H0 | 9310002 | 0.110672 | 0.108970 | 0.0000 | 0.0033 |
| E1H0 | 9310003 | 0.110623 | 0.104226 | 0.0000 | 0.0061 |
| E1H0 | 9310004 | 0.110988 | 0.107553 | 0.0000 | 0.0061 |
| E1H1 | 9310002 | 0.062747 | 0.055275 | 0.0000 | 0.0100 |
| E1H1 | 9310003 | 0.063020 | 0.053947 | 0.0000 | 0.0083 |
| E1H1 | 9310004 | 0.062057 | 0.055238 | 0.0006 | 0.0188 |

Hoffmann et al. (Nature Communications 2023, DOI 10.1038/s41467-023-42372-6) provide a Fig. 2h short-time mobility scale: source-data means are 0.0189192370 with SV and 0.0253090377 without SV, in $\mu m^2/s$. Source data: `https://doi.org/10.6084/m9.figshare.24179445.v1`. Their experiment selects condensate-core tracks at a 1-ms cadence, whereas the model analysis includes all Syn at a 0.14-ms saved cadence. Simulation positions have no localization error or exposure blur, and every model condition contains SV. Effective friction was chosen with reference to the literature before this comparison. The comparison provides a mobility scale reference, with higher absolute mobility in the model. Experimental uncertainty was not recalculated.

## S7. Reproduction and software checks

S1 Data contains full-precision inputs for all main figures, per-run structural and mobility values, shell comparisons and paired contrasts. The original 28-ms tables remain separate. The updated repository adds 12 common-84-ms analysis-input files, preserving all 600 frames and the exact stored values and types of the included arrays. Original raw outputs remain preserved separately. The portable script `analysis/common84ms.py` reproduces the new tables, and `scripts/render_common84ms.py` generates the four figures.

Raw-source hashes and exact checkpoint joins were checked, including positions, identities, radii and random-number state. Reconstruction of the earlier 2,400 frames reproduced the original structural and contact tables to absolute tolerance $10^{-12}$. Selected graph calculations were checked against all-pairs enumeration. Saved positions and radii use float32; one reconstructed overlap residual differed from the solver tolerance by $1.52 \times 10^{-6}$ nm, within its rounding bound. The recorded double-precision solver residual remained below 0.005 nm. Detailed numerical checks are retained with the analysis data.

## S8. Later assembly trends

For each trajectory, a straight line was fitted to $L$ within each 14-ms period. Entries are means of three per-run means and slopes. The periods below show the later development of each condition.

| Condition | Period (ms) | Mean $L$ | Slope ($ms^{-1}$) |
|---|---|---|---|
| E0H0 | 42–56 | 0.031581 | 0.000520 |
| E0H0 | 56–70 | 0.040241 | -0.000322 |
| E0H0 | 70–84 | 0.034406 | -0.000352 |
| E0H1 | 42–56 | 0.191023 | 0.003080 |
| E0H1 | 56–70 | 0.297030 | 0.008990 |
| E0H1 | 70–84 | 0.370037 | 0.005398 |
| E1H0 | 42–56 | 0.318683 | 0.007319 |
| E1H0 | 56–70 | 0.423941 | 0.006971 |
| E1H0 | 70–84 | 0.560191 | 0.006058 |
| E1H1 | 42–56 | 0.831795 | 0.008818 |
| E1H1 | 56–70 | 0.910580 | 0.001990 |
| E1H1 | 70–84 | 0.925225 | 0.000381 |